

Vision–Language Models

Understanding Image and Text

Image + Prompt → Text
Answer

3D Radiology
(CT / MRI)

Chest X-ray
(2D)

Pathology
(Whole-Slide)

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MedGemma 1.5 · MAIRA-2 · PathChat

From Retrieval to Generation

Section 2 matched images to text. A generative VLM goes further: it reads an image and a prompt and writes an answer. Three questions frame this session.

1

How does image information reach the LLM?

Encoders flatten pixels — 3D volumes, gigapixel slides — into tokens in one shared context window.

MedGemma 1.5

2

How does clinical context change the answer?

Priors, indication and history are fed with the image, shifting what the model concludes.

MAIRA-2 · PathChat

3

Is the claim supported by the image?

Grounding ties each sentence to a location; automated checks test every claim for truth and completeness.

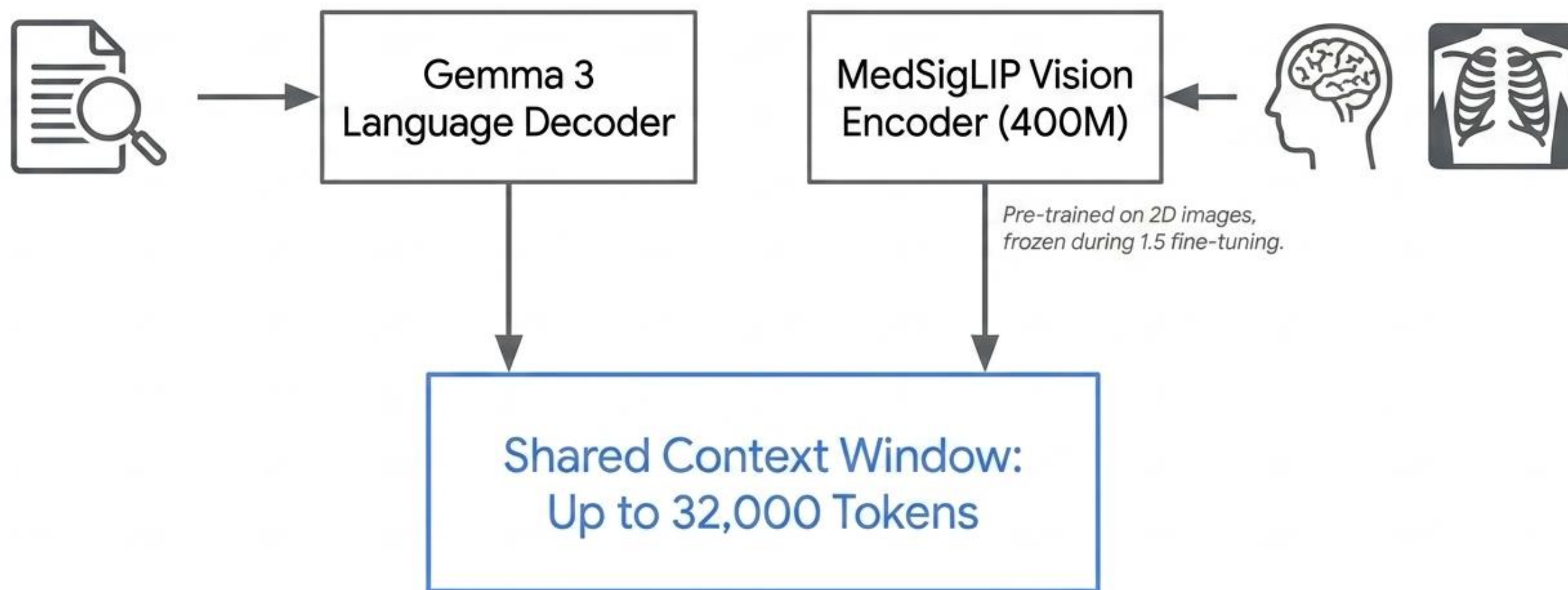
RadFact · Reliability

MedGemma 1.5

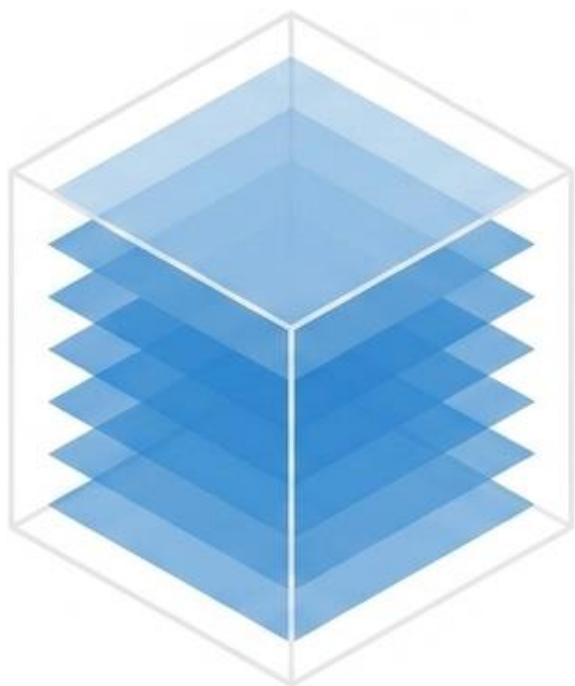
Multimodal Foundations for High-Dimensional Medical AI

4B Parameters | 32K Context

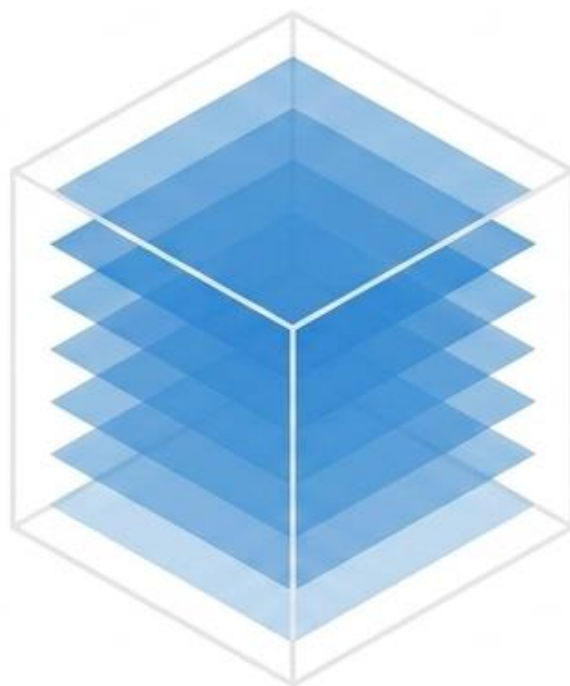
The Architecture of MedGemma 1.5



MedGemma 1.5 does not natively 'see' 3D or massive resolution files; it relies on novel preprocessing pipelines to flatten complex clinical data into a sequence of 2D vision tokens that fit within this 32K limit.



Raw Volumetric Data
(Varying slice counts
and resolutions).



Sample up to 85
equidistant axial slices.
Rescale each to
 896×896 pixels.



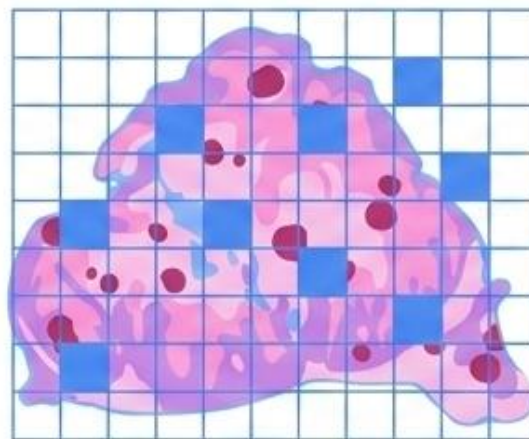
Maximum 21,760 Vision
Tokens per scan, leaving
remaining context for
clinical indication text.

Step 1: Low-Res Masking



Generate 5x magnification mask to isolate tissue-containing regions. Discard background.

Step 2: Stochastic Sampling



Stochastically sample 896x896 non-overlapping patches at 5x (34%), 10x (33%), and 20x (33%) magnifications.

Step 3: Sequential Capping



Extract maximum 126 patches per slide (32,256 tokens). Original spatial ordering is strictly preserved for relative context.

Prompt



You are analyzing contiguous CT slices. Is there evidence of a tumor in the liver? Explain your reasoning.

AI Response

...evidence of a large, irregular mass in the right lobe... heterogeneous enhancement... hyperdensity and hypodensity.

AI response evaluation by board-certified thoracic radiologist

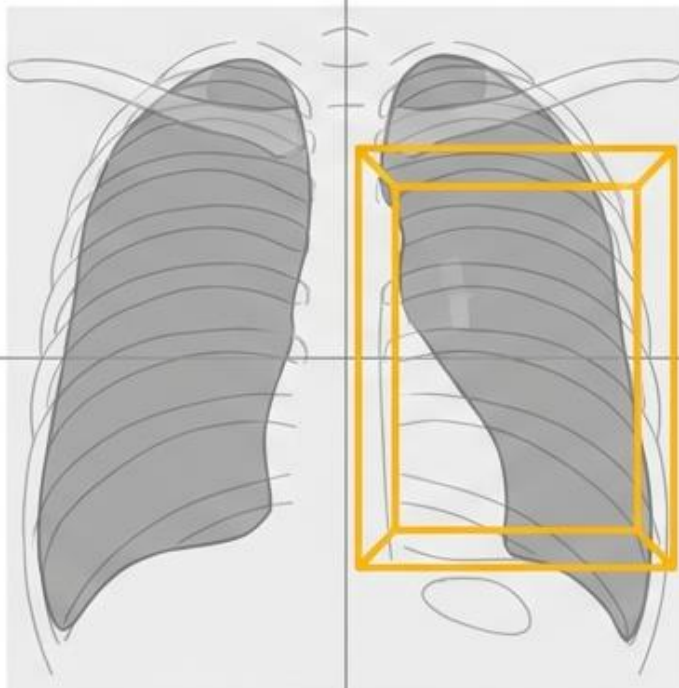


I agree there is a large tumor in the right hepatic lobe. I agree with the characterization of its heterogeneity, enhancement characteristics, and associated liver parenchymal distortion.



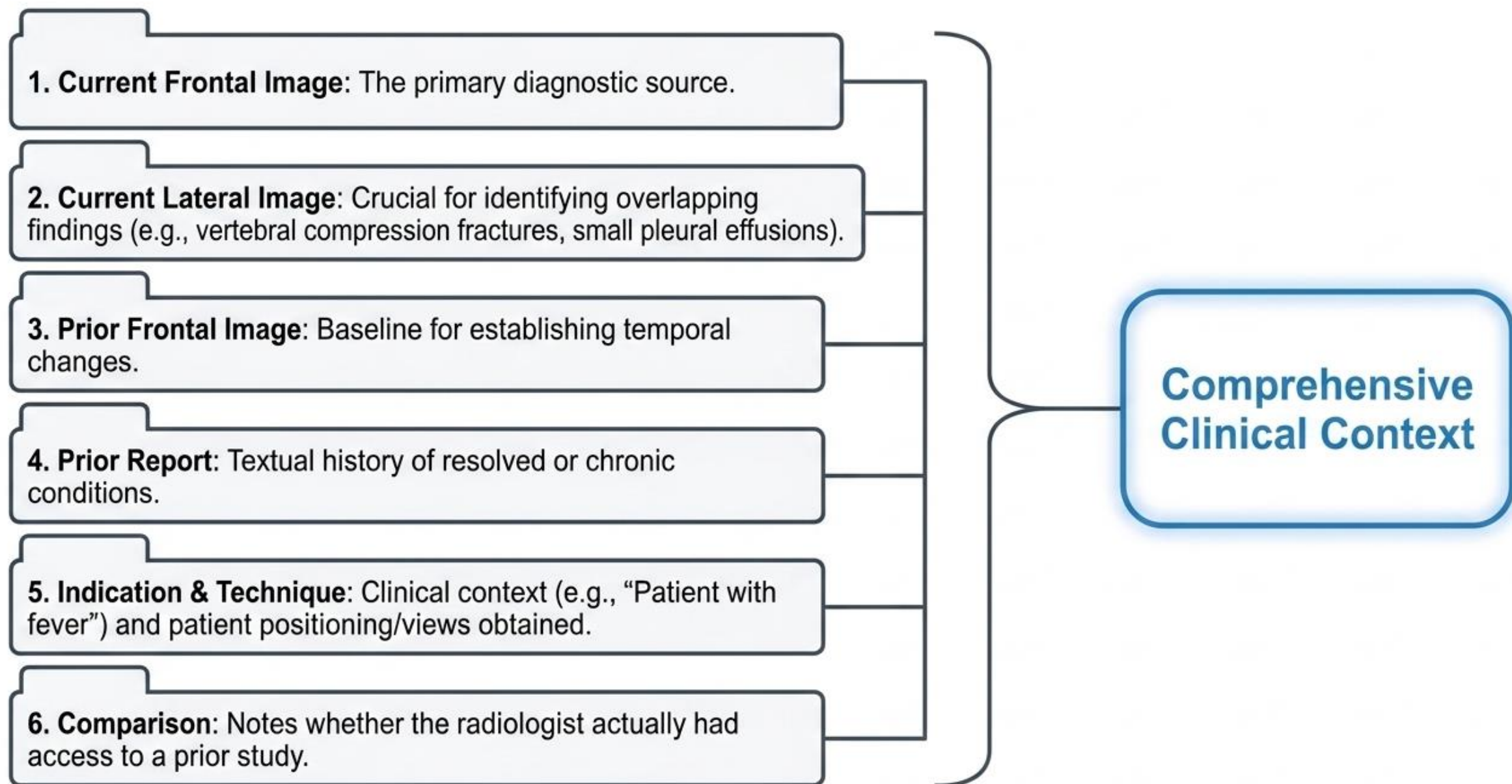
However, no substantial intrahepatic biliary ductal dilation is apparent.

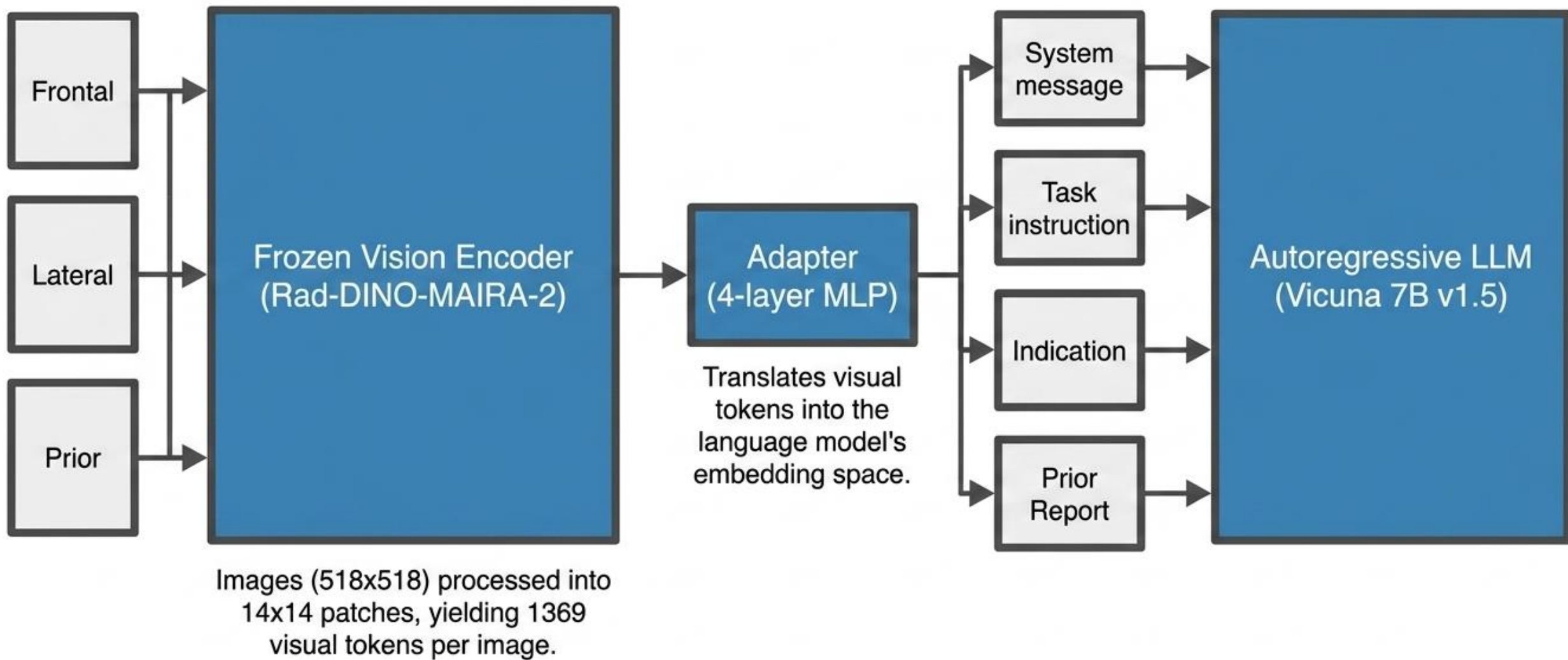
Grounded CXR Report Generation

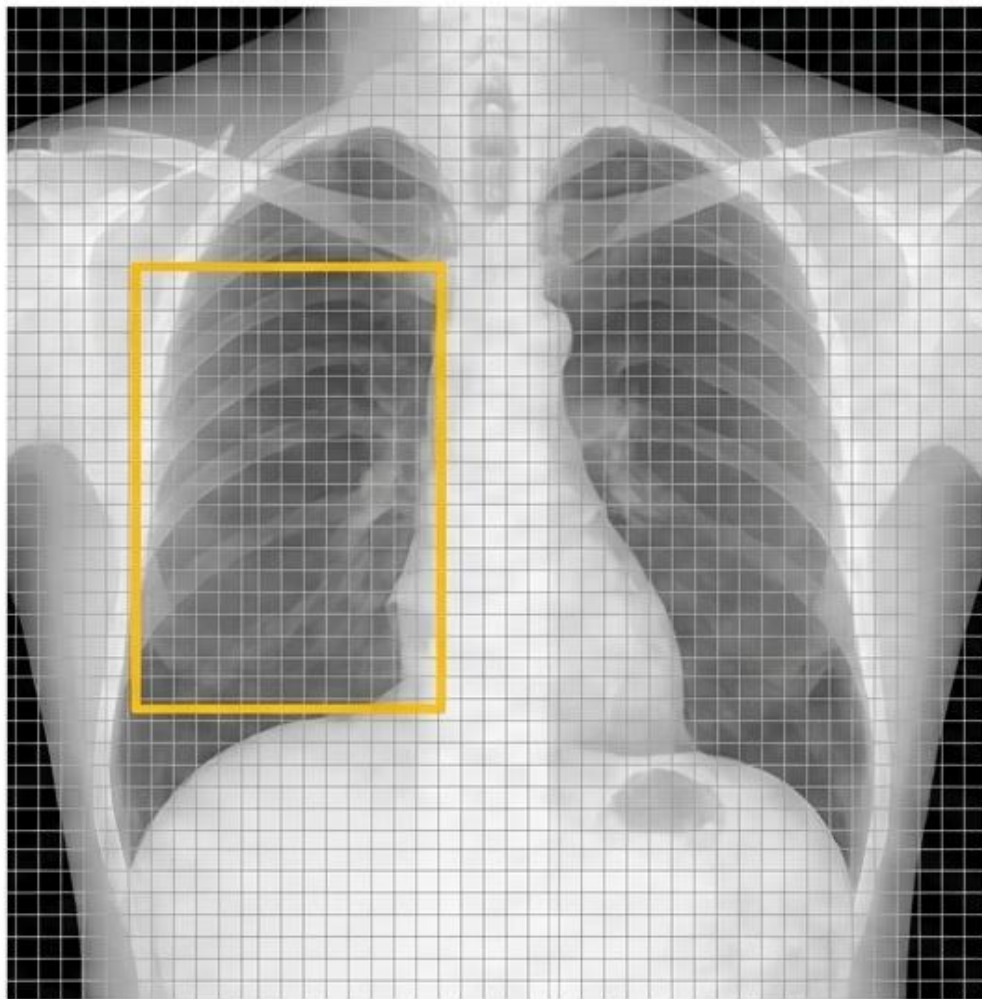


The Anatomy of a
Verifiable Medical AI

Deconstructing MAIRA-2,
Contextual Inputs, and the
RadFact Evaluation Framework.





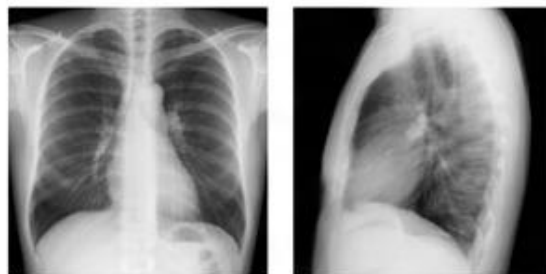


Disjoint Coordinate Tokenization (N=100 grid)

```
<obj>  
  "There is a left-sided pneumothorax."  
  <box><x12><y23><x34><y45></box>  
</obj>
```

The LLM vocabulary is expanded with specialized coordinate tokens. The model generates bounding boxes by outputting tokens for the top-left (**x1**, **y1**) and bottom-right (**x2**, **y2**) coordinates within `<box>` delimiters, intrinsically linking the generated text sentence to a 2D spatial location.

Input & Output



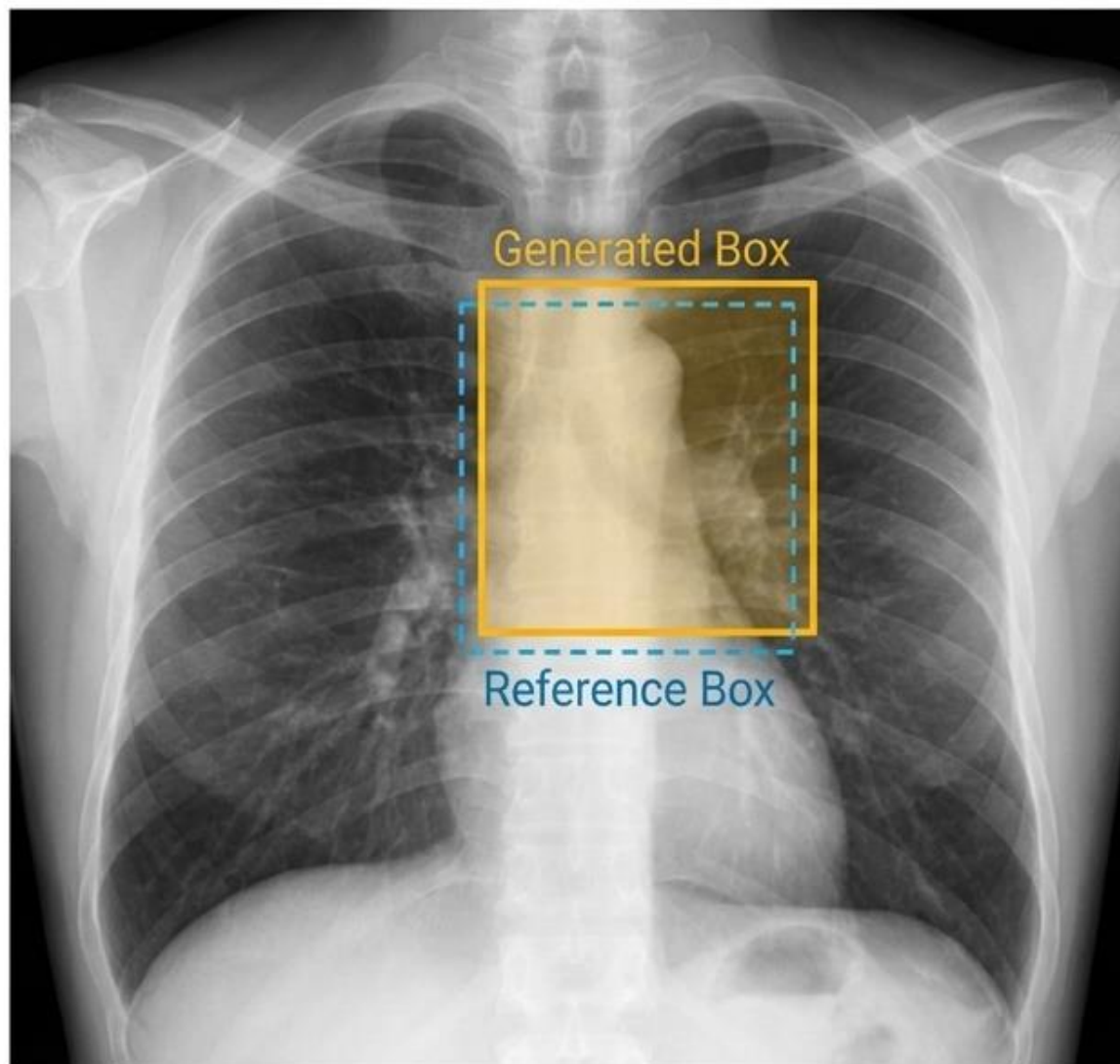
Frontal

Lateral

Generated Text:

There is uncoiling of the aorta compatible with hypertension.

Grounding Alignment



RadFact Scoring

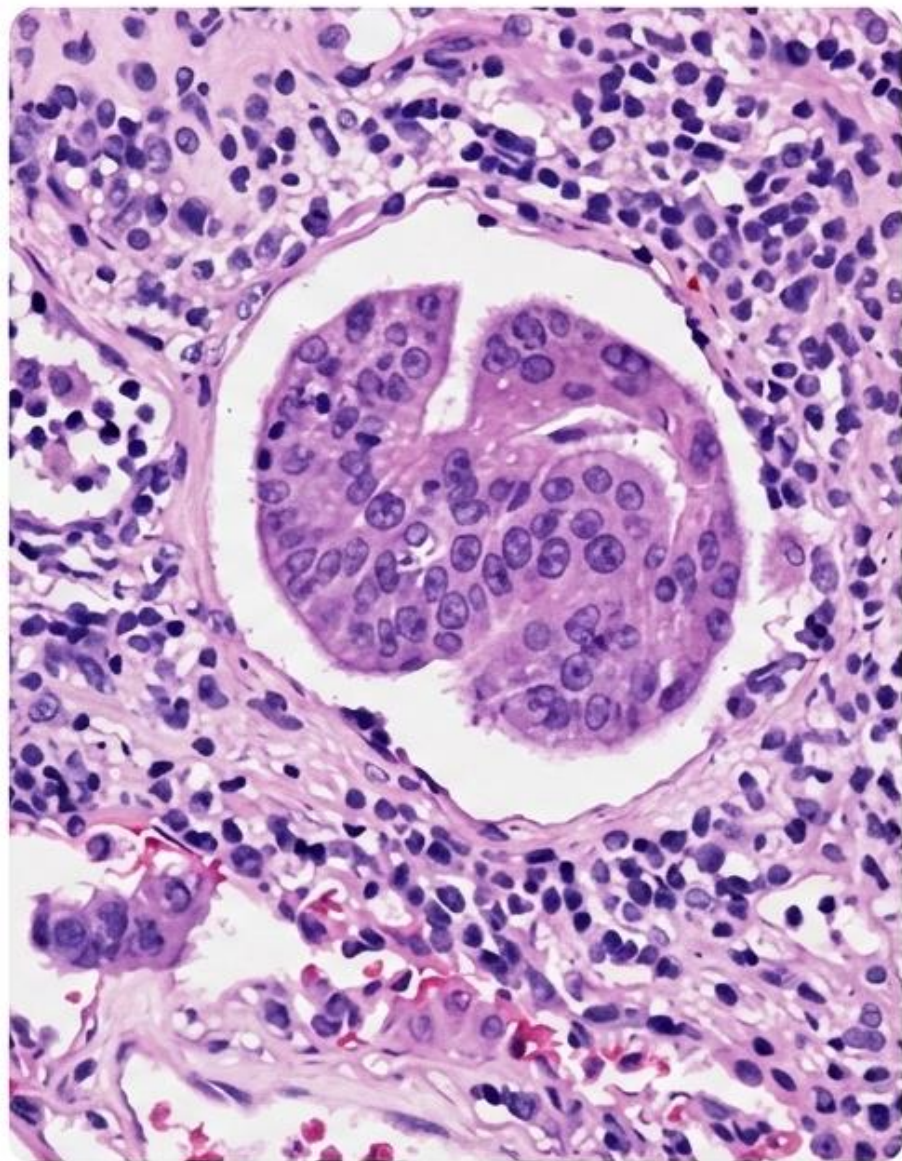
Logical Precision

Check: Entailed
Entailed (Matches reference text: 'Uncoiling of the aorta.')

Grounding Precision

Check: 1.0 (Boxes overlap sufficiently).

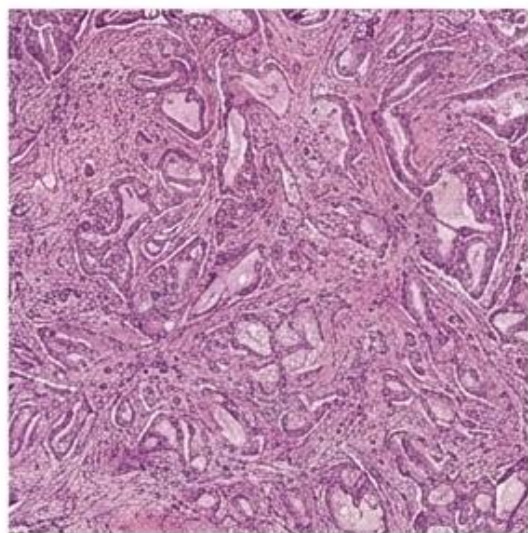
Result: The finding is verified both textually and spatially.



PathChat: A Multimodal Generative AI Copilot for Human Pathology

Based on research from Nature, 2024 by Lu, Chen, Williamson et al.

Combining visual features with clinical history improves diagnostic accuracy by over 11%.



Visual Features

[+]

63-year-old male presents with chronic cough and unintentional weight loss. Chest X-ray shows a 3 cm mass.

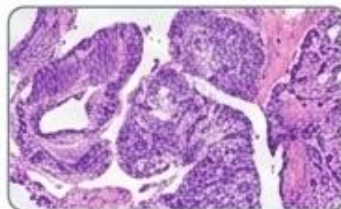
Clinical Text

[=]

Diagnosis: Lung Adenocarcinoma.

When provided only the text context without the image, performance drops substantially, proving PathChat relies heavily on actual visual interpretation.

Interactive multi-turn conversations enable AI-assisted differential diagnosis



Initial Diagnosis Request

AI Assessment: Suggests primary lung adenocarcinoma vs squamous cell carcinoma. Recommends TTF-1, Napsin A, and p40 stains.

Human Input: TTF-1, p40, and synaptophysin stains were negative. What next?

AI Refinement: Tumor is likely not of lung origin. Recommends CK7, CK20, and CDX2 stains to identify a metastatic primary site.

Capability vs. Reliability

Evaluating Multimodal AI in Medicine

A critical analysis of SOTA performance, the strict hallucination penalty, and the blindspots in automated fact-checking.

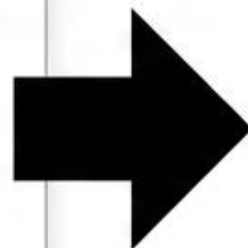
The Hallucination Penalty: Zero Tolerance

The image demonstrates a high-grade brain tumor. ✓ Correct

There is a notable presence of numerous mitotic figures. ✓ Correct

The tumor cells are strongly positive for estrogen receptors.

✗ Hallucination

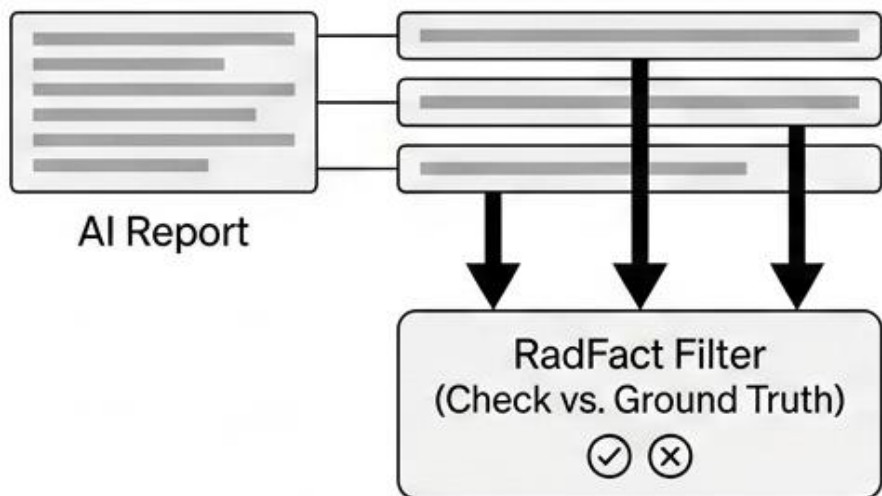


**SCORE: 0% /
INCORRECT**

Note: In PathChat's evaluation, a response was graded as entirely incorrect if ANY portion was hallucinated, regardless of surrounding accuracy.

The Automated Alternative: RadFact

How it works

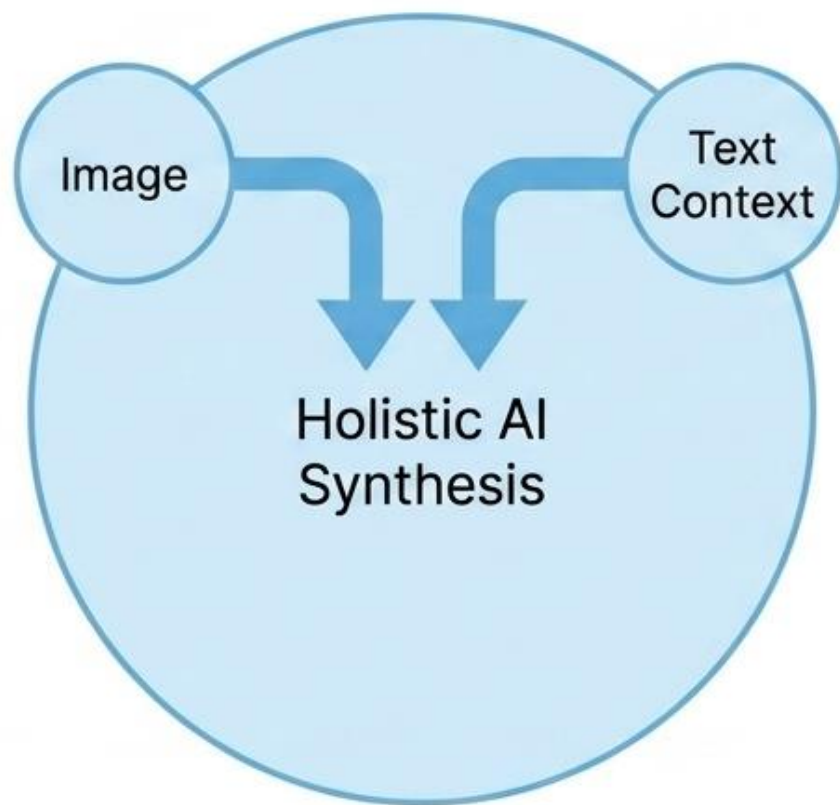


Evaluates factuality against
a reference report.

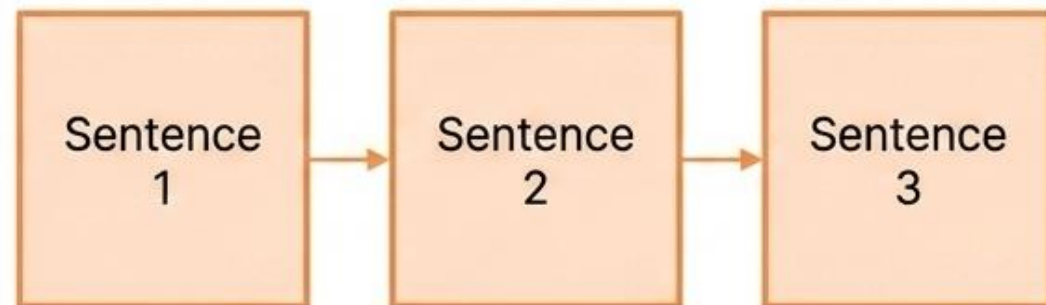
Sentence-Level Analysis

RadFact evaluates the factuality of a report one sentence at a time. It never compares generated sentences to one another.

The Capability vs. Reliability Gap



State-of-the-art models
operate holistically.



Current automated evaluators operate sequentially,
rendering them blind to structural logic errors.

Next-Generation Evaluation

We have built multimodal AI capable of synthesizing complex medical truths. Our next mandate is building automated evaluators capable of understanding them.

